

CELEBAL DATA ENGINEERING INTERNSHIP

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WEEK 4 Assignment

Objective: To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Task 1: Explore Azure Portal & Create Resource Group

Do:

- Explore Azure Portal
- Create a Resource Group

Deliverable: Screenshot of Resource Group

Created resource group 'celebal-internship-cei' in Central India under Azure for Students subscription.

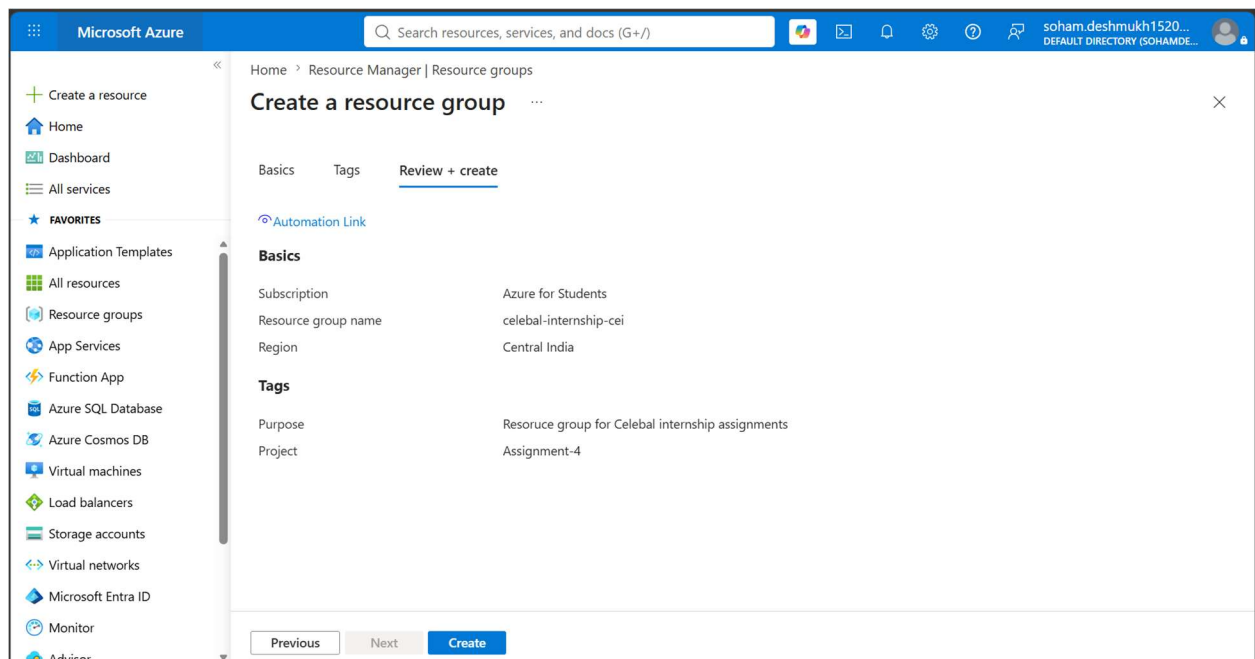


Figure: Resource Group creation review

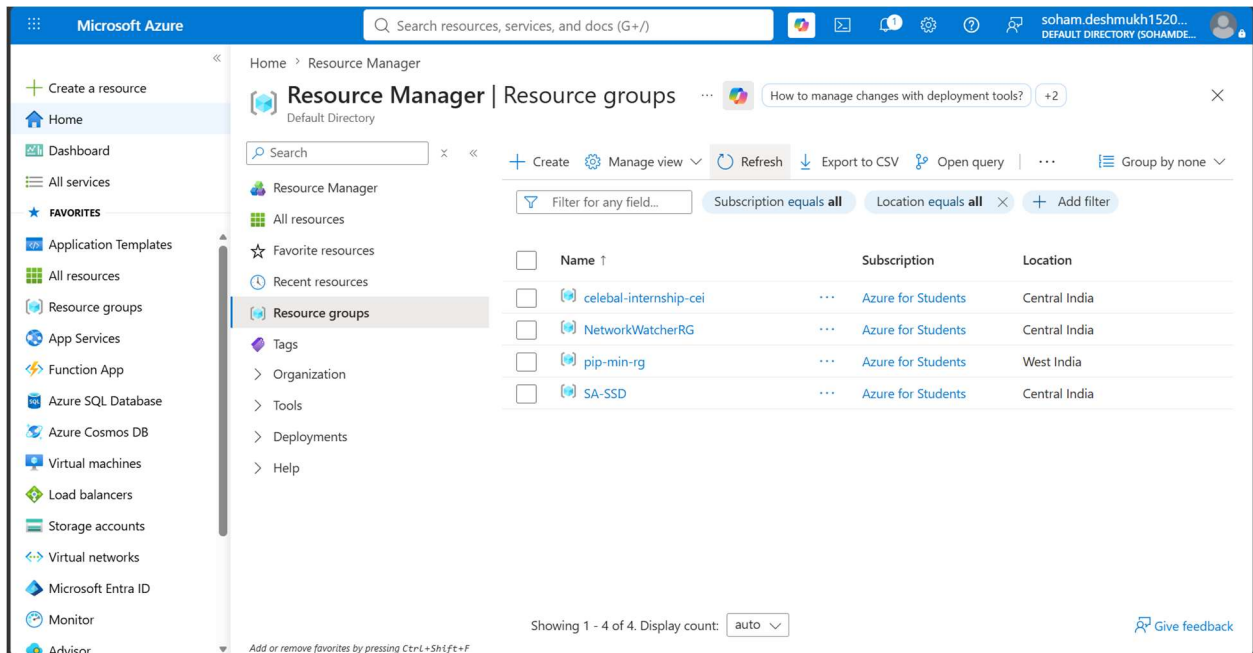


Figure: Resource Groups list

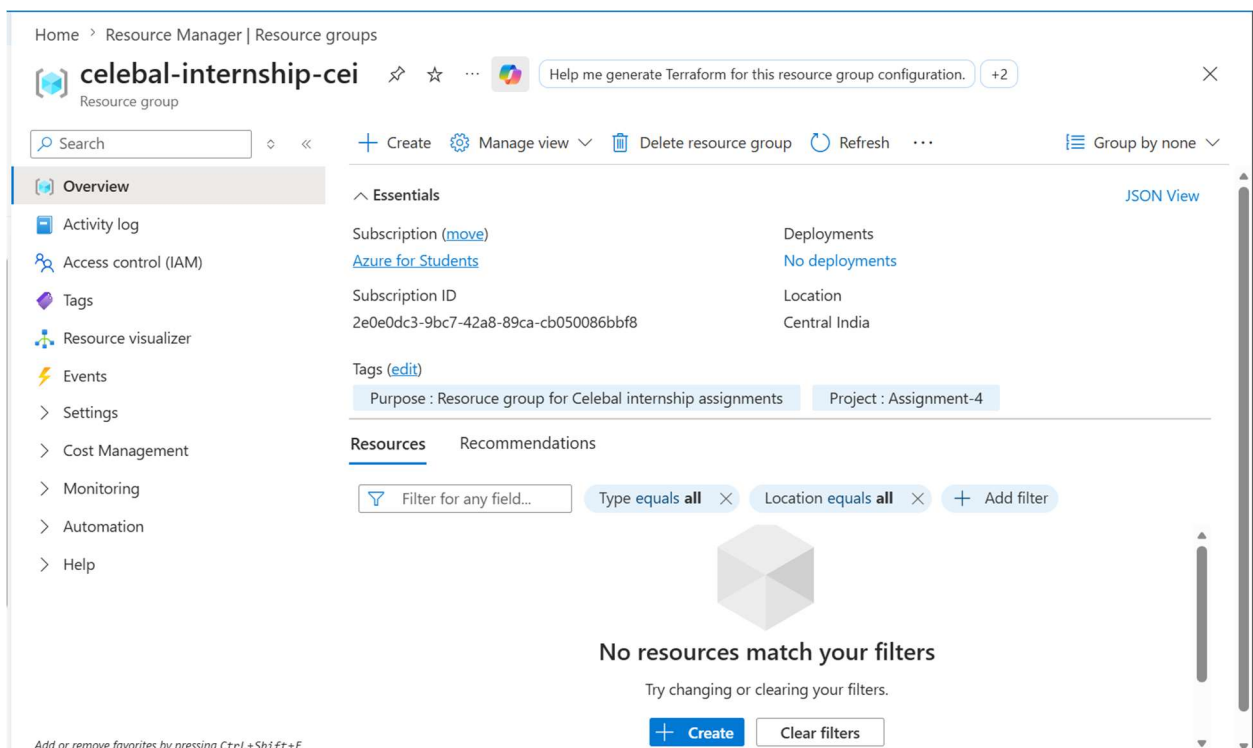


Figure: Created Resource Group overview

Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable: Screenshot of container with uploaded CSV

Storage account and blob container were configured successfully. Sample-Superstore.csv uploaded.

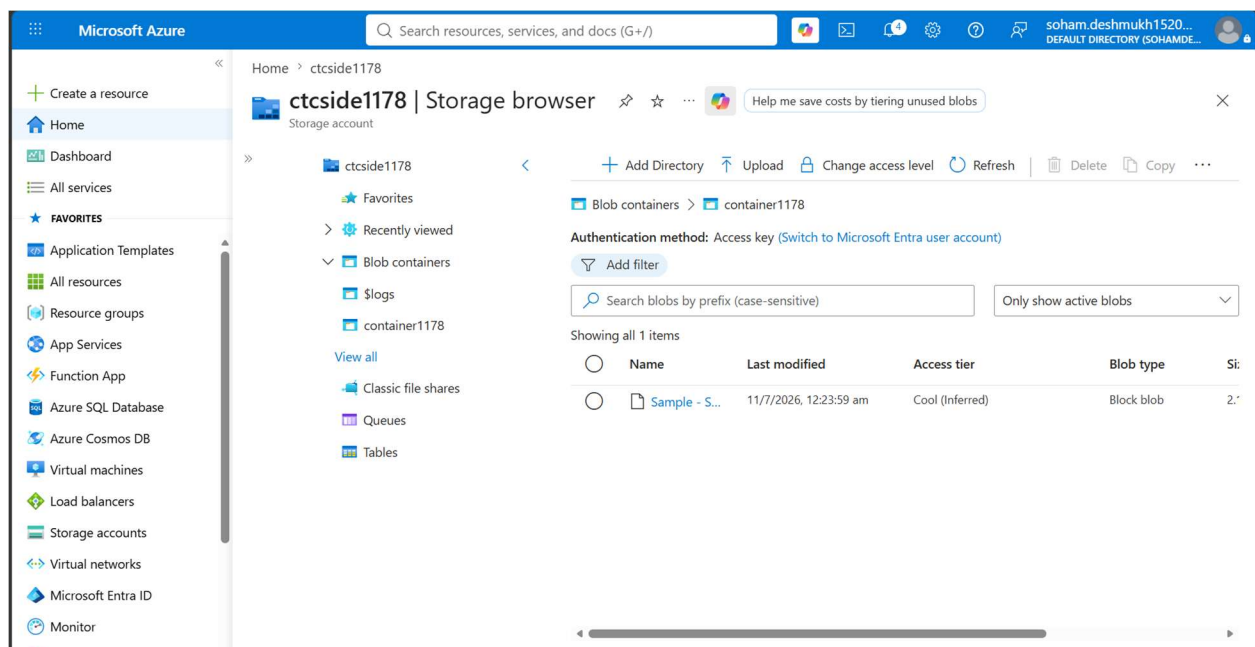


Figure: Blob container with uploaded CSV

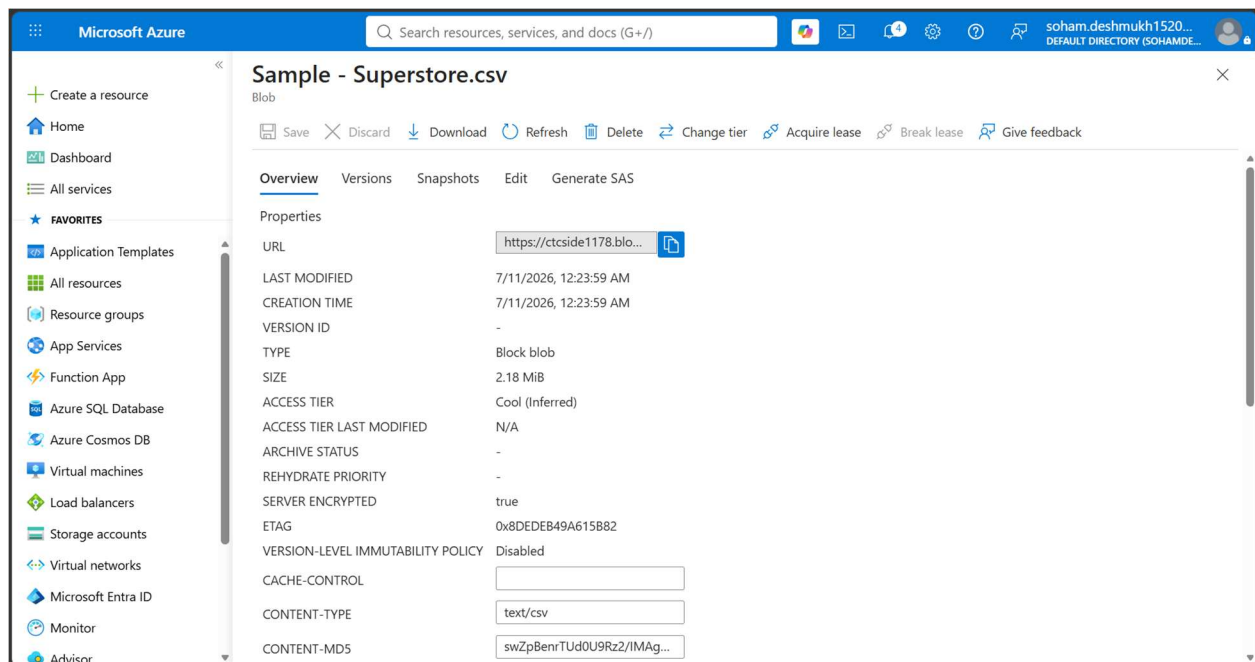


Figure: Uploaded CSV blob properties

Task 3: Azure Data Factory Basics

Do:

- Create Azure Data Factory
- Explore Author, Monitor and Manage
- Create Linked Service
- Create Source and Destination Datasets
- Configure Get Metadata

Deliverable: Screenshots of Linked Service, Datasets and Get Metadata activity

Linked service AzureBlobStorage1 was created to connect ADF with Azure Blob Storage. Two datasets (source and destination) were configured. Get Metadata validates file existence, size and last modified time.

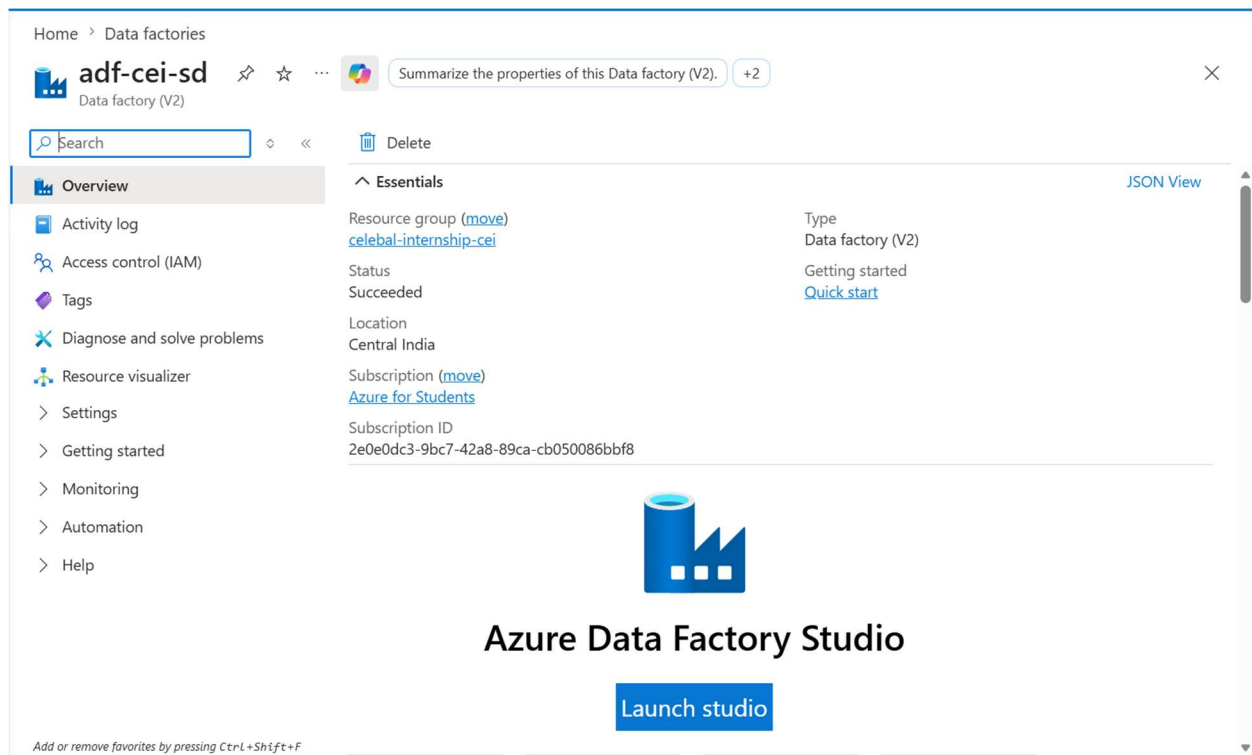


Figure: Azure Data Factory overview

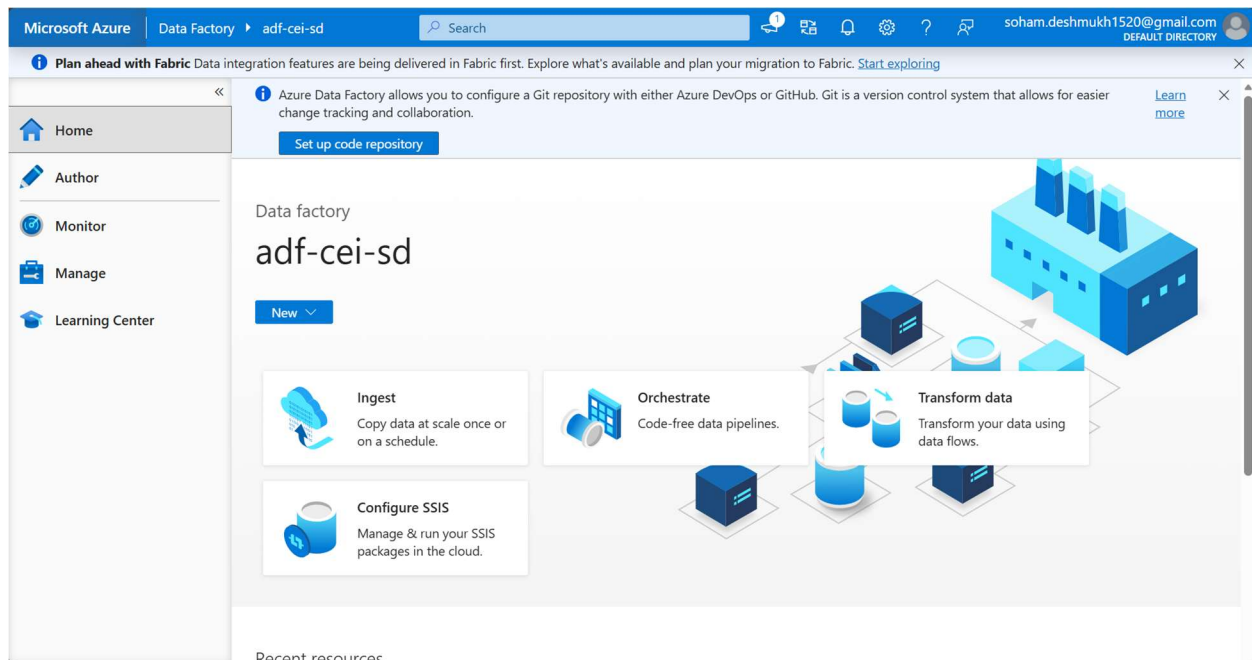


Figure: ADF Studio

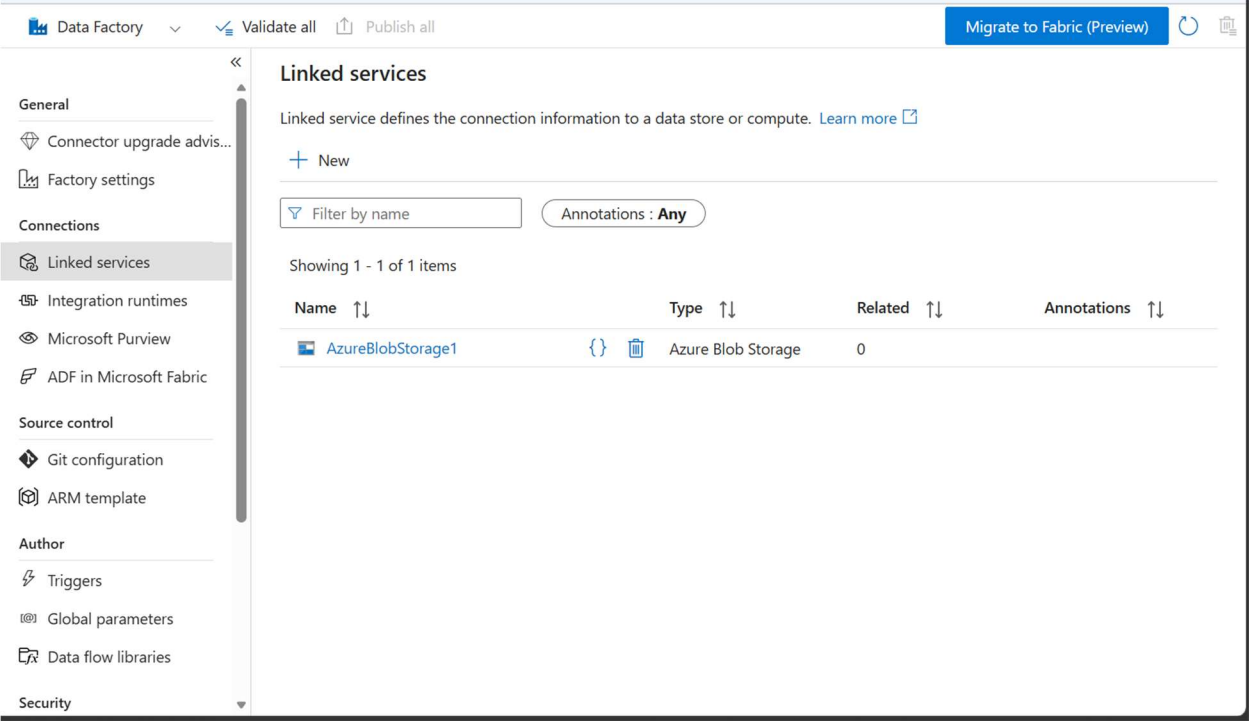


Figure: Linked Service

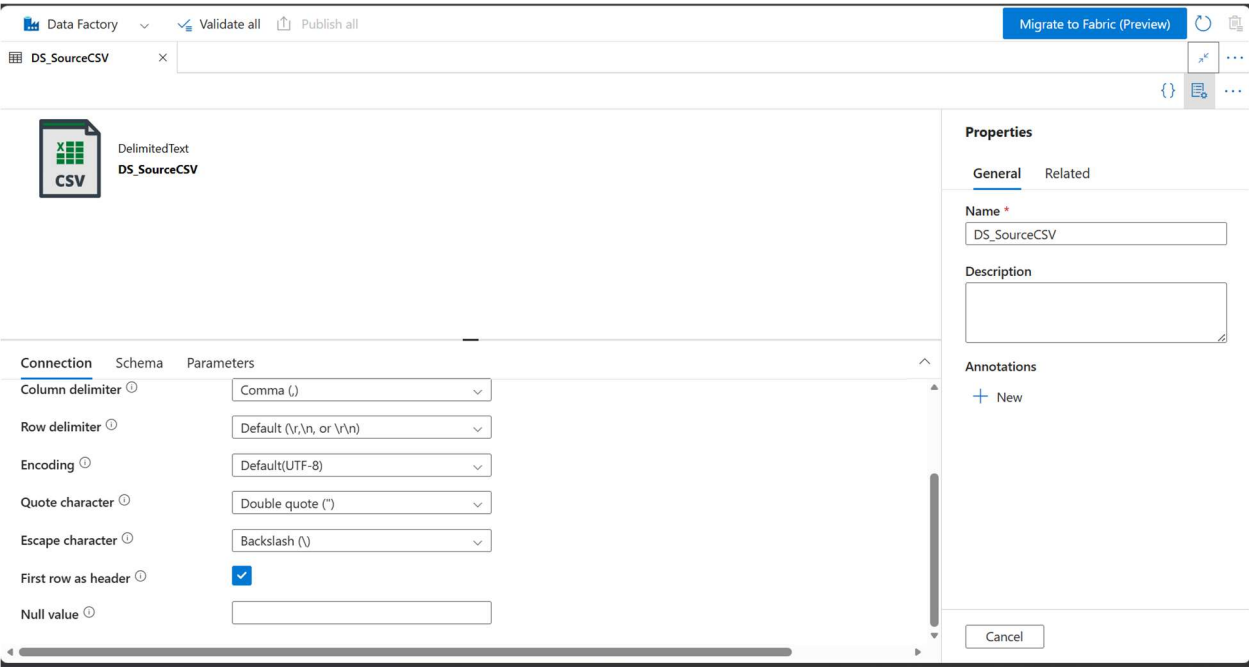


Figure: Source Dataset

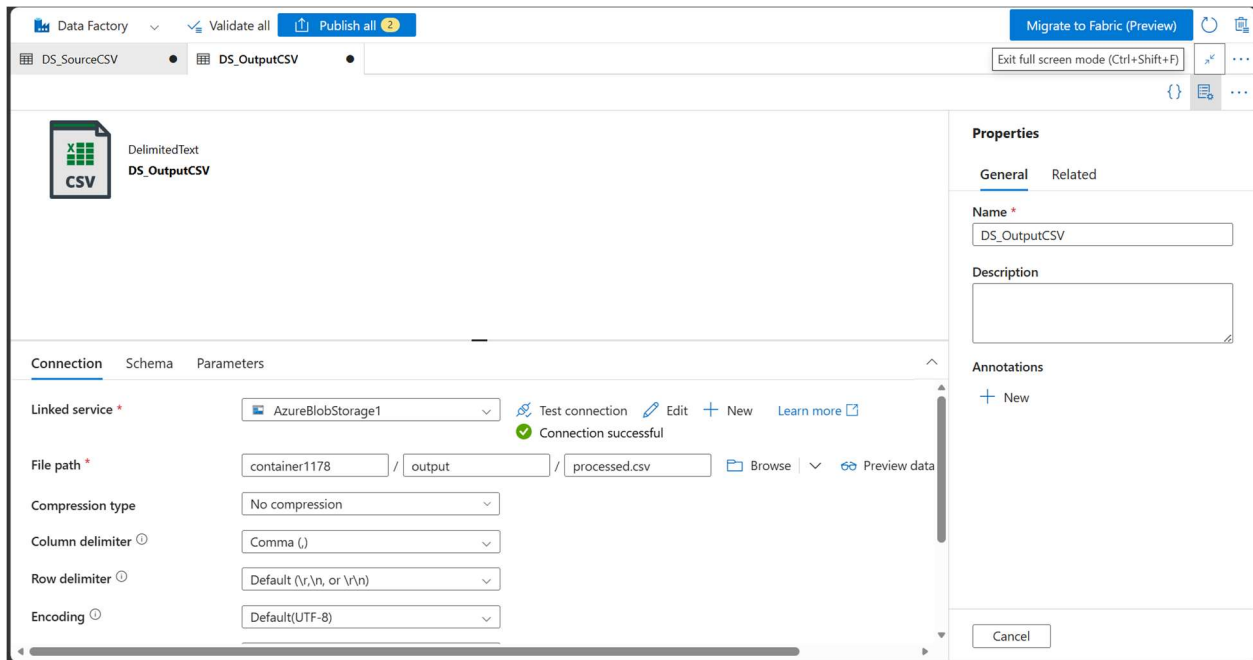


Figure: Destination Dataset

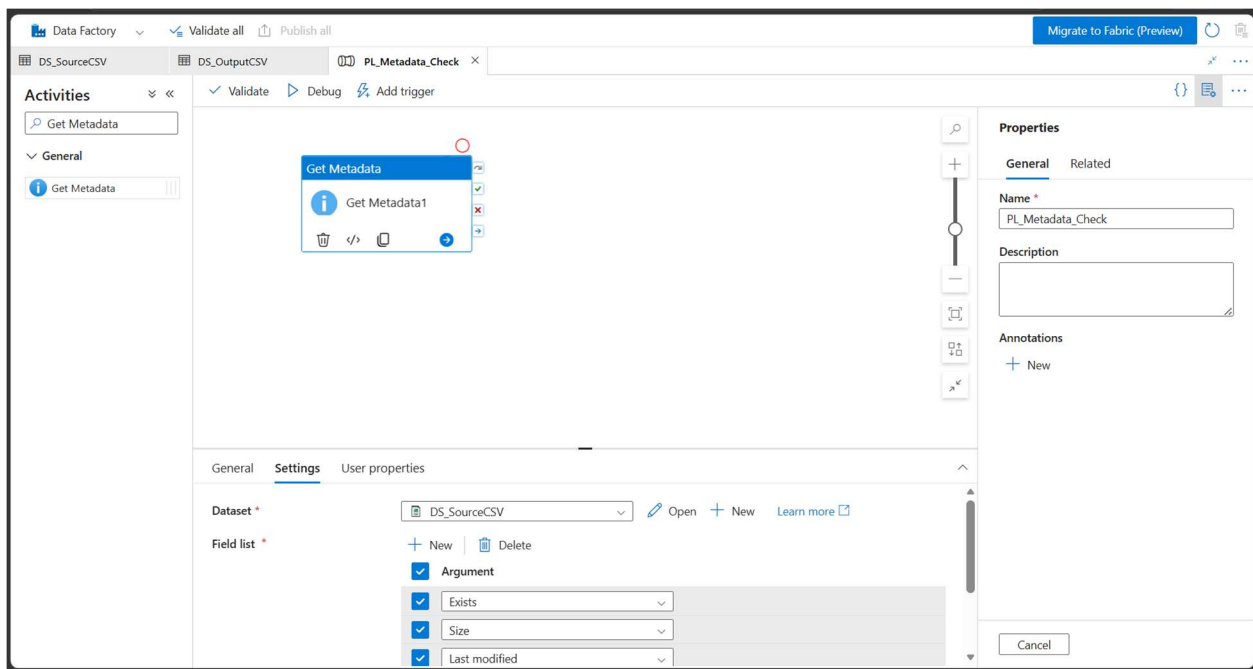


Figure: Get Metadata activity

Task 4: Pipeline Development

Do:

- Create Copy Data pipeline
- Configure Source and Sink datasets
- Connect Get Metadata to Copy Data
- Validate pipeline

Deliverable: Screenshot of pipeline design

ForEach activity was optional and not required because only a single CSV file was processed.

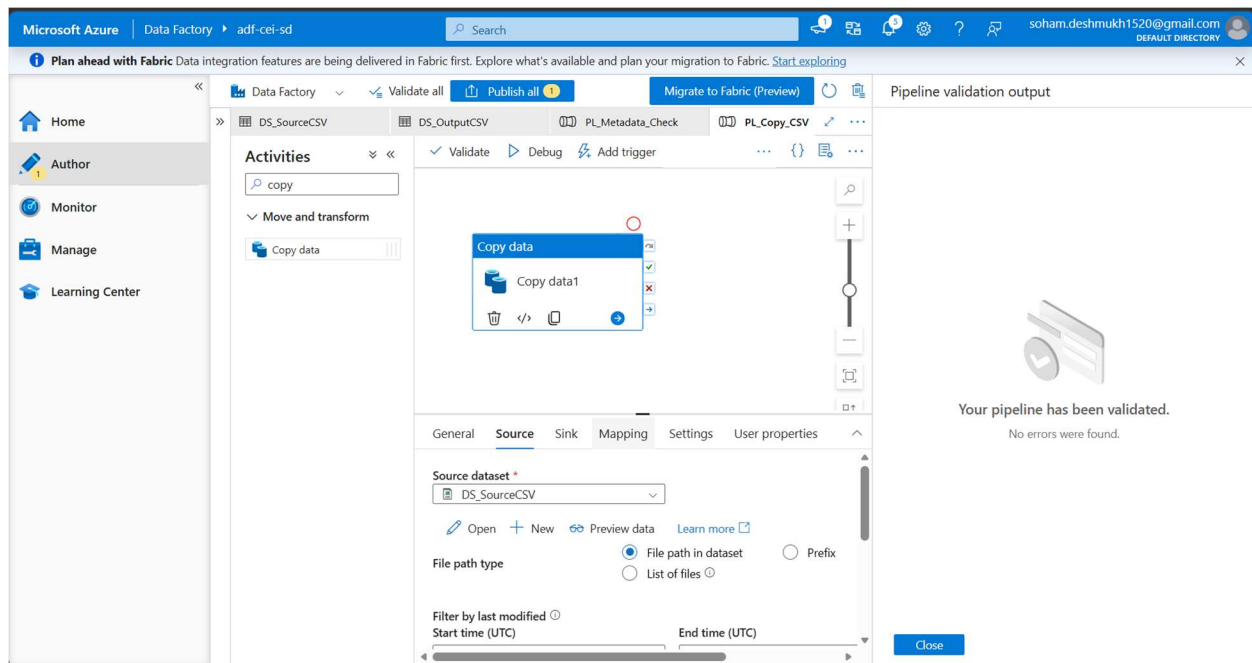


Figure: Copy Data activity configuration and design

Task 5: Pipeline Execution

Do:

- Run pipeline using Debug
- Verify successful execution
- Validate copied output

Deliverable: Pipeline execution succeeded

Pipeline executed successfully through Debug mode. Metadata validation completed and the CSV file was copied to output/processed.csv.

The screenshot displays the Microsoft Azure Data Factory console. The top navigation bar includes the 'Data Factory' breadcrumb and a search bar. The left sidebar contains navigation links for Home, Author, Monitor, Manage, and Learning Center. The main workspace shows a pipeline named 'PL_Copy_CSV' with a 'Copy data' activity. The 'Output' tab is selected, showing the pipeline run ID 'c7708e88-0b01-40fa-bdb6-6c6a840fa46b' and a status of 'Succeeded'. Below this, a table lists the activities and their execution details.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	Succeeded	Copy data	7/12/2026, 5:54:37 PM	15s	AutoResolveIntegrationRuntime (Ce

Figure: Copy Data succeeded

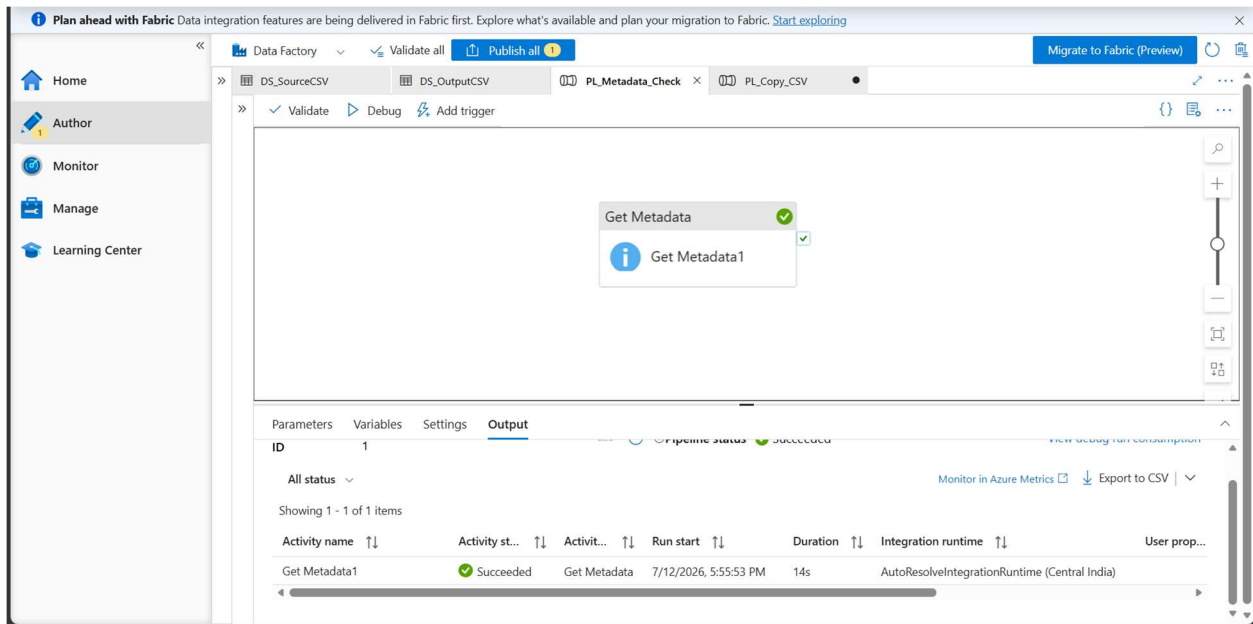


Figure: Get Metadata succeeded

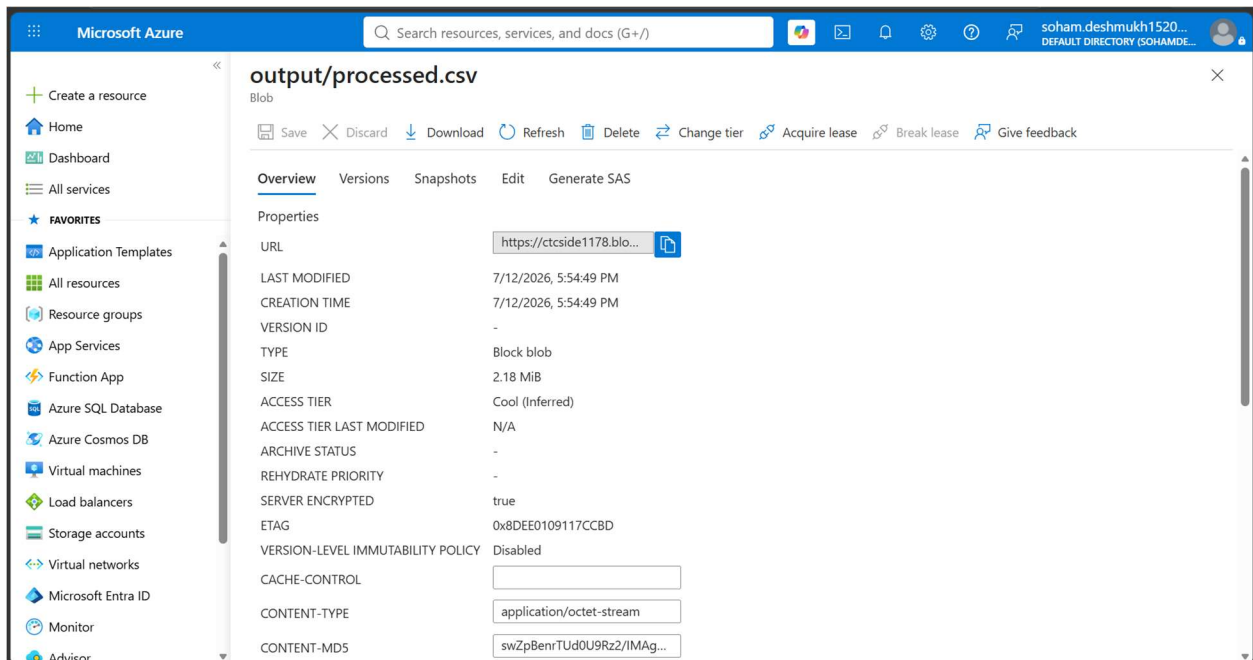


Figure: Processed CSV blob

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount
1	CA-2016-1	6/16/2016	6/16/2016	Standard	Cl CG-12520	Claire Gut Consumer	United Sta	Henderson	Kentucky	42420	South	FUR-BO-1	Furniture	Bookcases	Bush Soma	261.96	2	
2	CA-2016-1	6/16/2016	6/16/2016	Standard	Cl CG-12520	Claire Gut Consumer	United Sta	Henderson	Kentucky	42420	South	FUR-CH-1	Furniture	Chairs	Hon Delux	731.94	3	
3	CA-2016-1	6/16/2016	6/16/2016	Standard	Cl DV-13045	Darrin Van Corporate	United Sta	Los Angeles	California	90036	West	OFF-LA-1	Office Sup	Labels	Self-Adhes	14.62	2	
4	US-2015-1	10/18/201	10/18/201	Standard	CSO-20335	Sean O'Do Consumer	United Sta	Fort Lauderdale	Florida	33311	South	FUR-TA-1	Furniture	Tables	Bretford C	957.5775	5	
5	US-2015-1	10/18/201	10/18/201	Standard	CSO-20335	Sean O'Do Consumer	United Sta	Fort Lauderdale	Florida	33311	South	OFF-ST-10	Office Sup	Storage	Eldon Fold	22.368	2	
6	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc Consumer	United Sta	Los Angeles	California	90032	West	FUR-FU-1	Furniture	Furnishing	Eldon Expr	48.86	7	
7	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc Consumer	United Sta	Los Angeles	California	90032	West	OFF-AR-1	Office Sup	Art	Newell 322	7.28	4	
8	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc Consumer	United Sta	Los Angeles	California	90032	West	TEC-PH-1	Technolog	Phones	Mitel 5320	907.152	6	
9	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc Consumer	United Sta	Los Angeles	California	90032	West	OFF-BI-10	Office Sup	Binders	DXL Angle	18.504	3	
10	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc Consumer	United Sta	Los Angeles	California	90032	West	OFF-AP-1	Office Sup	Appliance	Belkin F5C	114.9	5	
11	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc Consumer	United Sta	Los Angeles	California	90032	West	FUR-TA-1	Furniture	Tables	Chromcra	1706.184	9	
12	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc Consumer	United Sta	Los Angeles	California	90032	West	TEC-PH-1	Technolog	Phones	Konftel 25i	911.424	4	
13	CA-2017-14/15/2017	4/20/2017	4/20/2017	Standard	CAA-10480	Andrew All Consumer	United Sta	Concord	North Carolina	28027	South	OFF-PA-1	Office Sup	Paper	Xerox 1967	15.552	3	
14	CA-2016-1	6/16/2016	6/16/2016	Standard	CIM-15070	Irene Mad Consumer	United Sta	Seattle	Washington	98103	West	OFF-BI-10	Office Sup	Binders	Fellowes P	407.976	3	
15	US-2015-11/22/201	11/26/201	11/26/201	Standard	CHP-14815	Harold Pav Home Offi	United Sta	Fort Worth	Texas	76106	Central	OFF-AP-1	Office Sup	Appliance	Holmes Re	68.81	5	
16	US-2015-11/22/201	11/26/201	11/26/201	Standard	CHP-14815	Harold Pav Home Offi	United Sta	Fort Worth	Texas	76106	Central	OFF-BI-10	Office Sup	Binders	Storax Dur	2.544	3	
17	CA-2014-1	6/14/2014	6/14/2014	Standard	CPK-19075	Pete Kriz Consumer	United Sta	Madison	Wisconsin	53711	Central	OFF-ST-10	Office Sup	Storage	Stur-D-Sto	665.88	6	
18	CA-2014-15/13/2014	5/15/2014	5/15/2014	Second	Cl AG-10270	Alejandro Consumer	United Sta	West Jord	Utah	84084	West	OFF-ST-10	Office Sup	Storage	Fellowes S	55.5	2	
19	CA-2014-18/27/2014	6/14/2014	6/14/2014	Standard	Cl ZD-21925	Zuschuss Consumer	United Sta	San Francisco	California	94109	West	OFF-AR-1	Office Sup	Art	Newell 341	8.56	2	
20	CA-2014-18/27/2014	6/14/2014	6/14/2014	Standard	Cl ZD-21925	Zuschuss Consumer	United Sta	San Francisco	California	94109	West	TEC-PH-1	Technolog	Phones	Cisco SPA	213.48	3	
21	CA-2014-18/27/2014	6/14/2014	6/14/2014	Standard	Cl ZD-21925	Zuschuss Consumer	United Sta	San Francisco	California	94109	West	OFF-BI-10	Office Sup	Binders	Wilson Jon	22.72	4	
22	CA-2014-18/27/2014	6/14/2014	6/14/2014	Standard	Cl ZD-21925	Zuschuss Consumer	United Sta	San Francisco	California	94109	West	OFF-BI-10	Office Sup	Binders	Wilson Jon	22.72	4	

Figure: Original CSV preview

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount
1	CA-2016-1	6/16/2016	6/16/2016	Standard	Cl CG-12520	Claire Gut Consumer	United Sta	Henderson	Kentucky	42420	South	FUR-BO-1	Furniture	Bookcases	Bush Soma	261.96	2	
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8	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc Consumer	United Sta	Los Angeles	California	90032	West	TEC-PH-1	Technolog	Phones	Mitel 5320	907.152	6	
9	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc Consumer	United Sta	Los Angeles	California	90032	West	OFF-BI-10	Office Sup	Binders	DXL Angle	18.504	3	
10	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc Consumer	United Sta	Los Angeles	California	90032	West	OFF-AP-1	Office Sup	Appliance	Belkin F5C	114.9	5	
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13	CA-2017-14/15/2017	4/20/2017	4/20/2017	Standard	CAA-10480	Andrew All Consumer	United Sta	Concord	North Carolina	28027	South	OFF-PA-1	Office Sup	Paper	Xerox 1967	15.552	3	
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15	US-2015-11/22/201	11/26/201	11/26/201	Standard	CHP-14815	Harold Pav Home Offi	United Sta	Fort Worth	Texas	76106	Central	OFF-AP-1	Office Sup	Appliance	Holmes Re	68.81	5	
16	US-2015-11/22/201	11/26/201	11/26/201	Standard	CHP-14815	Harold Pav Home Offi	United Sta	Fort Worth	Texas	76106	Central	OFF-BI-10	Office Sup	Binders	Storax Dur	2.544	3	
17	CA-2014-1	6/14/2014	6/14/2014	Standard	CPK-19075	Pete Kriz Consumer	United Sta	Madison	Wisconsin	53711	Central	OFF-ST-10	Office Sup	Storage	Stur-D-Sto	665.88	6	
18	CA-2014-15/13/2014	5/15/2014	5/15/2014	Second	Cl AG-10270	Alejandro Consumer	United Sta	West Jord	Utah	84084	West	OFF-ST-10	Office Sup	Storage	Fellowes S	55.5	2	
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20	CA-2014-18/27/2014	6/14/2014	6/14/2014	Standard	Cl ZD-21925	Zuschuss Consumer	United Sta	San Francisco	California	94109	West	TEC-PH-1	Technolog	Phones	Cisco SPA	213.48	3	
21	CA-2014-18/27/2014	6/14/2014	6/14/2014	Standard	Cl ZD-21925	Zuschuss Consumer	United Sta	San Francisco	California	94109	West	OFF-BI-10	Office Sup	Binders	Wilson Jon	22.72	4	
22	CA-2014-18/27/2014	6/14/2014	6/14/2014	Standard	Cl ZD-21925	Zuschuss Consumer	United Sta	San Francisco	California	94109	West	OFF-BI-10	Office Sup	Binders	Wilson Jon	22.72	4	

Figure: Processed CSV preview

Task 6: IAM Roles

Do:

- Assign Reader role
- Assign Contributor/Data Factory Contributor role
- Grant Storage Blob Data Contributor access to ADF managed identity

Deliverable: Screenshot of role assignments

ADF managed identity was granted Storage Blob Data Contributor access to interact with Azure Blob Storage securely.

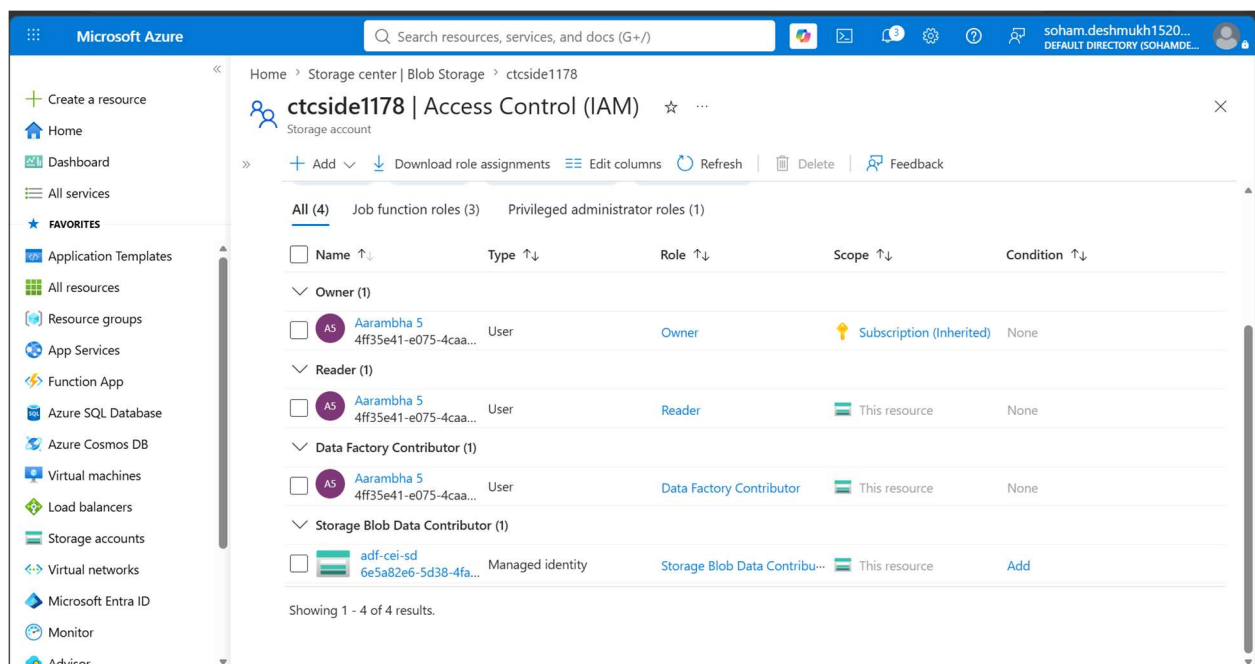


Figure: Storage Blob Data Contributor assigned to ADF managed identity

Mini Project (Final Task)

Problem Statement: Build a complete pipeline that reads a CSV file from Azure Blob Storage, validates metadata using Get Metadata, and copies the CSV to a new location in Blob Storage.

Implementation Summary

- Source: Sample-Superstore.csv stored in Azure Blob Storage.
- Linked Service: AzureBlobStorage1.
- Datasets: DS_SourceCSV and DS_OutputCSV.
- Metadata Validation: Get Metadata activity checks file existence, size and last modified date.
- Pipeline: PL_CSV_DataPipeline executes Get Metadata followed by Copy Data.
- Destination: output/processed.csv in the same blob container.
- Result: Pipeline executed successfully and destination file matched the source data.

Microsoft Azure | Data Factory | adf-cei-sd | Search | soham.deshmukh1520@gmail.com | DEFAULT DIRECTORY

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Home | Author | Monitor | Manage | Learning Center

Dashboards | Runs | Pipeline runs | Trigger runs | Change Data Capture (previ...) | Runtimes & sessions | Integration runtimes | Data flow debug | Notifications | Alerts & metrics

All pipeline runs > PL_CSV_DataPipeline - Activity runs

Rerun | Cancel | Refresh | Update pipeline | List | Gantt

Get Metadata (Check_Source_File) → Copy data (Copy_CSV_File)

Activity runs

Pipeline run ID: 6f30083a-3598-4dc2-b335-c87cf3c88cef

All status | Monitor in Azure Metrics | Export to CSV

Showing 1 - 2 items

Activity name	Activity st...	Activity type	Run start	Duration	Integration runtime
Copy_CSV_File	Succeeded	Copy data	7/12/2026, 6:25:27 PM	15s	AutoResolveIntegrationRuntime (Cr
Check_Source_File	Succeeded	Get Metadata	7/12/2026, 6:25:11 PM	16s	AutoResolveIntegrationRuntime (Cr

Figure: Final Mini Project Pipeline Execution

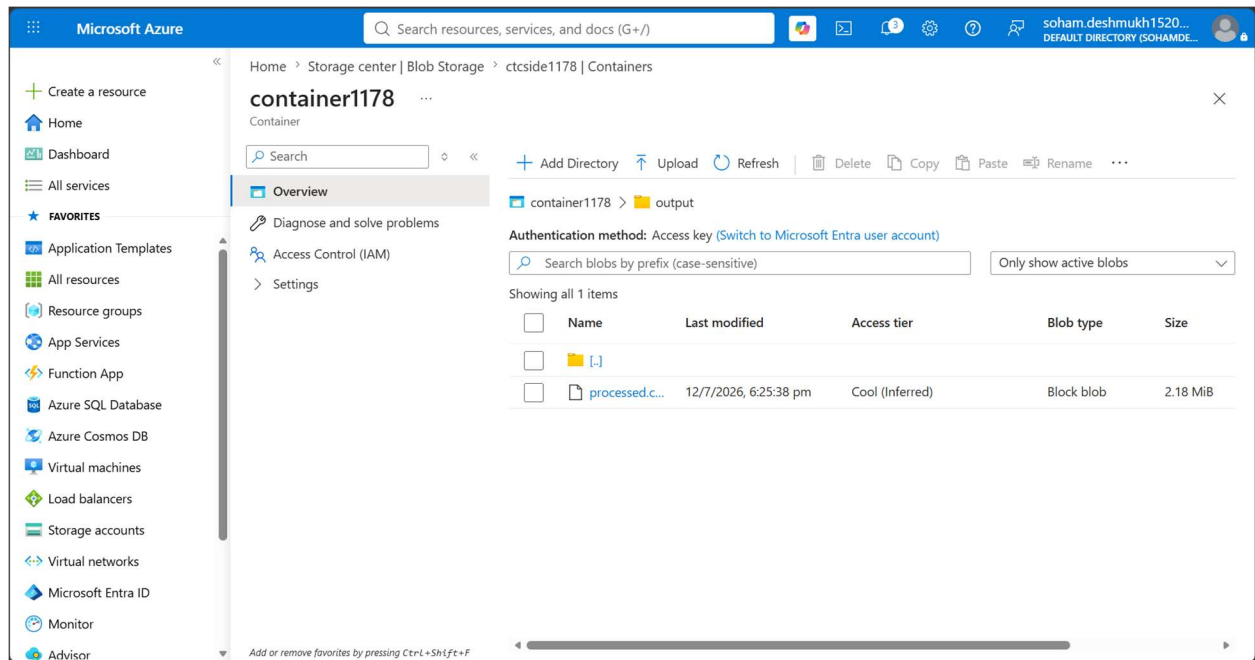


Figure: Data copied to destination

Input given:

```
{
  "dataset": {
    "referenceName": "DS_SourceCSV",
    "type": "DatasetReference",
    "parameters": {}
  },
  "fieldList": [
    "exists",
    "size",
    "lastModified"
  ],
  "storeSettings": {
    "type": "AzureBlobStorageReadSettings",
    "enablePartitionDiscovery": false
  },
  "formatSettings": {
    "type": "DelimitedTextReadSettings"
  }
}
```

Output

Copy to clipboard

```
{
  "exists": true,
  "size": 2287806,
  "lastModified": "2026-07-10T18:53:59Z",
  "effectiveIntegrationRuntime":
    "AutoResolveIntegrationRuntime (Central India)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 11
  }
}
```

Monitor in Azure Metrics

Export to CSV

Run start	Duration	Integration runtime
7/12/2026, 6:25:27 PM	15s	AutoResolveIntegrationRuntime (C...
7/12/2026, 6:25:11 PM	16s	AutoResolveIntegrationRuntime (C...

Check_Source_File

Succeeded

Get Metadata

Figure: Meta Data Verified